Stafford's identity over simple Noetherian rings with infinite centre

Christopher Albert[1*], Johanna Moser[1] and Maximilian Philipp[1]

[1*]Graz University of Technology, Graz, Austria.

*Corresponding author(s). E-mail(s): albert@tugraz.at;

Abstract

Let $\boldsymbol{R}$ be a simple right Noetherian ring with infinite centre and let $\boldsymbol{d} \in \boldsymbol{R}$ be a regular element such that $\boldsymbol{R}/\boldsymbol{dR}$ has finite Krull dimension. We prove that for every nonzero $\boldsymbol{a} \in \boldsymbol{R}$ and every $\boldsymbol{B} \in \boldsymbol{R}$ there are $\boldsymbol{H}, \boldsymbol{U}, \boldsymbol{V} \in \boldsymbol{R}$ with $\mathbf{1} = \boldsymbol{dU} + (\boldsymbol{B} + \boldsymbol{Ha})\boldsymbol{dV}$. For $\boldsymbol{a} = \mathbf{1}$ and $\boldsymbol{B} = \mathbf{0}$ this is the identity $\boldsymbol{R} = \boldsymbol{dR} + \boldsymbol{HdR}$ that Stafford conjectured in 1978 for the Weyl algebras. If $\boldsymbol{R}/\boldsymbol{dR}$ has finite Krull dimension for every regular $\boldsymbol{d}$, for instance if $\mathbf{Kdim}\, \boldsymbol{R} \leq \boldsymbol{\omega}$, then every finitely generated Goldie-torsion right $\boldsymbol{R}$-module is cyclic and every right ideal of $\boldsymbol{R}$ is generated by two elements. Hence Stafford's conjecture holds for the Weyl algebras in characteristic zero (for nonzero $\boldsymbol{d}$), finitely generated torsion modules over them are cyclic, as conjectured in Bellamy's survey, and one-sided ideals of the ring of differential operators on a smooth irreducible affine variety in characteristic zero are generated by two elements, which answers a question of Björk. The results extend to simple filtered algebras over infinite fields whose associated graded ring is commutative Noetherian of finite Krull dimension, and to simple quotients of enveloping algebras of finite-dimensional Lie algebras over infinite fields. The proof is a downward induction on Krull dimension that applies the inductive hypothesis twice to the same right ideal and combines the two answers by a generic central scalar. All results are formalized in Lean.

Keywords: simple Noetherian ring, Krull dimension, number of generators, torsion module, Weyl algebra, ring of differential operators

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1 Introduction

Stafford proved that every right ideal of the Weyl algebra $A_n = A_n(k)$ over a field k of characteristic zero is generated by two elements [1, Thm. 3.1], and that finitely generated torsion left A_n-modules are stably two-generated [1, Thm. 3.7]. Here, following [1, p. 435], a right module M is *stably m-generated* if for every $r \geq m$ and all $a_1, \dots, a_{r+1} \in M$ with $M = \sum_{i=1}^{r+1} a_i R$ there are $f_1, \dots, f_r \in R$ with $M = \sum_{i=1}^{r} (a_i + a_{r+1} f_i) R$, and *stably cyclic* if it is stably 1-generated; for left modules the definition is symmetric. He conjectured the following.

Conjecture 3.8 of [1] *Let $d \in A_n$. Then there exists $f \in A_n$ such that $A_n = dA_n + fdA_n$.*

The statement does not exclude $d = 0$, for which it fails; Stafford's argument for the equivalence below uses only nonzero d, and we read the conjecture for nonzero d. Stafford noted that the conjecture is true if $n = 1$ and that it is equivalent to every finitely generated torsion right A_n-module being stably cyclic [1, p. 438].

We prove the identity of Conjecture 3.8, in the stronger form $1 = dU + (B + Ha)dV$ with a prescribed nonzero right factor a, for every regular d in a simple right Noetherian ring R with infinite centre for which R/dR has finite Krull dimension (Theorem A). This covers $A_n(k)$ for $\operatorname{char} k = 0$, so Conjecture 3.8 holds for every nonzero d. When R/dR has finite Krull dimension for every regular d, every finitely generated Goldie-torsion module is cyclic and every right ideal is generated by two elements (Corollary C). Consequently every one-sided ideal of the ring $D(X)$ of differential operators on a smooth irreducible affine variety X in characteristic zero is generated by two elements, which answers Björk's question [2, Problem 2.10], and finitely generated torsion $A_n(k)$-modules are cyclic, which is Conjecture 3.4 of Bellamy's survey [3] (Theorem 7.4).

Throughout, modules are *right* modules and ideals without qualification are right ideals. An element of a ring R is *regular* if it is neither a left nor a right zero-divisor. A module M is *Goldie-torsion* if every $m \in M$ satisfies $ms = 0$ for some regular $s \in R$. We write $\operatorname{Kdim} M$ for the Krull dimension of M in the sense of Gabriel and Rentschler [4], that is, the deviation of the lattice of submodules of M [5, 6.1.2, 6.2.2]; the zero module has Krull dimension -1 (Section 2). We use the following hypotheses on a ring R.

- (S) R is simple;
- (N) R is right Noetherian;
- (Z) the centre $k = Z(R)$ is infinite;
- ($\mathrm{H_{all}}$) $\operatorname{Kdim}(R/dR)$ is finite for every regular $d \in R$;
- (H_ω) $\operatorname{Kdim} R_R \leq \omega$.

Under (S) the centre is a field, so (Z) says that R is an algebra over an infinite central field. By "$\operatorname{Kdim} M$ is finite" we mean that $\operatorname{Kdim} M \leq n$ for some integer n. For a right Noetherian ring, (H_ω) implies ($\mathrm{H_{all}}$) (Lemma 2.3).

Theorem A *Assume* (S), (N) *and* (Z). *Let* $d \in R$ *be regular with* $\operatorname{Kdim}(R/dR)$ *finite. Then for every nonzero* $a \in R$, *which may be a zero-divisor, and every* $B \in R$ *there are* $H, U, V \in R$ *with*

$$1 = dU + (B + Ha)dV.$$

In particular $R = dR + HdR$ *for some* $H \in R$.

Theorem A is the case $I = dR$, $x = Bd$, $t = ad$ of the following.

Theorem A′ *Assume* (S), (N) *and* (Z). *Let* I *be a right ideal of* R *that contains a regular element and for which* $\operatorname{Kdim}(R/I)$ *is finite. Then for every* $x \in R$ *and every nonzero* $t \in R$ *there is* $H \in R$ *with*

$$I + (x + Ht)R = R.$$

Corollary B *Assume* (S), (N), (Z) *and* (H_ω). *Then the conclusion of Theorem A holds for every regular* $d \in R$, *and that of Theorem A′ for every right ideal that contains a regular element.*

Corollary C *Assume* (S), (N), (Z) *and* ($\mathrm{H}_{\mathrm{all}}$), *for instance* ($\mathrm{H}_\omega$).

(C.1) *Every finitely generated Goldie-torsion module* M *is cyclic. More precisely, for every* $x \in M$ *and every nonzero* $t \in R$ *there is* $h \in M$ *with* $M = (x + ht)R$.

(C.2) *Every right ideal* I *is generated by two elements. More precisely, there is* $a \in I$ *such that every* $y \in I$ *has a regular* s *with* $ys \in aR$; *and for every such* a, *every* $y \in I$ *and every nonzero* $t \in R$ *there is* $h \in I$ *with* $I = aR + (y + ht)R$.

Part (C.1) holds whenever $\operatorname{Kdim} M$ is finite, and it has a stable form (Proposition 6.1). Corollary C applied to R^{op} gives the statements for left ideals and left modules whenever R^{op} satisfies the hypotheses, as it does for the rings of Section 7 (Theorem 7.4).

Context

For commutative Noetherian rings the number of generators of a module is bounded by local data plus the dimension [6, 7]; Eisenbud and Evans obtained such generators by elementary transformations [8], and Swan's lemma [7, Lemma 4], which adds to a generator a multiple of another by an arbitrary central unit, is a commutative model of the form $x + Ht$. These bounds were extended by Warfield to fully bounded rings [9, 10], by Stafford to Noetherian rings [11] and by Coutinho to right Noetherian rings of finite Krull dimension [12]. Right ideals of a simple right Noetherian ring of Krull dimension n are generated by $n + 1$ elements [13, Cor. 1.5], a bound Robson derived from cyclic objects in the quotient of the module category by modules of smaller Krull dimension [14, Def. 1.4, Thm. 4.3], using that such quotients of finitely generated modules have finite length, for which he refers to Gordon and Robson [15]. For $\operatorname{Kdim} R = 1$, finitely generated torsion modules over a simple Noetherian ring are cyclic (a result of Robson, see [10, Cor. 7.1]), and R is *very simple* [16, Prop. 1.3]: it is

prime and Noetherian, and for all $a, b, c, d \in R$ with d regular there are $f, g \in R$ with $aR + bR + cR = (a + cfd)R + (b + cgd)R$ [16, p. 419]. Right ideals of a very simple domain are stably 2-generated [16, Prop. 1.5(a)]; Bell applied this to simple domains of quadratic growth [17]. When R/dR has finite length and R is simple Noetherian, Theorem A follows from [18, Prop. 1.3], which Stafford presents there as a special case of [19, Thm. 5.1], a theorem he attributes to Robson [19, p. 232]. Lemma 4.3 for a single test is close to the left-right mirror of [19, Thm. 5.1], which is stated there for simple left Noetherian rings and has no hypothesis on the centre; it lowers the Krull dimension once, with a prescribed factor, and each further drop costs a new generator [19, Thm. 5.5]. For $D(X)$, Coutinho and Holland proved that right ideals are stably 3-generated [16, Thm. 2.3, Cor. 2.4] and conjectured two-generation [16, Conj. 4.3], following Björk [2, Problem 2.10]. Caro and Levcovitz, and Caro-Tuesta and Levcovitz, proved two-generation for differential operators over fields of Laurent series [20, Thm. 5.2], [21, Thm. 1], and refer to two-generation of one-sided ideals of every simple Noetherian ring as a conjecture [20]; Stafford had considered such an extension unlikely [1, §5, Question 4]. Bellamy's survey records the cyclicity of torsion A_n-modules as open [3, Conj. 3.4], recalls Björk's question [3, §6] and asks whether the results of [1] extend to simple quantizations [3, §7.1]. For simple filtered algebras over a field k of characteristic zero, of infinite dimension, with finite-dimensional filtration pieces and $F_0 = k$, whose associated graded ring is a commutative affine domain of finite Krull dimension, the identity of Theorem A, for all nonzero d and a and with all factors in the algebra, was formalized earlier by a different route [22].

The method

The proof of Theorem A′ is a downward induction on e, where the inductive statement asks for an element $f = x + Ht$ with $\operatorname{Kdim} R/(I + fR) < e$ that in addition satisfies a finite list of *tests*: conditions saying that certain germs, the morphisms modulo modules of Krull dimension below a given level, are epic (Sections 3 and 4). A generic central scalar satisfies finitely many tests at once. Each step of the induction uses the inductive hypothesis twice for the *same* ideal I: once to obtain f, and once more to obtain a correction g that removes the top-dimensional part of $R/(I + fR)$ while keeping $\operatorname{Kdim} R/(I + gbR) \leq e$, where b is regular with $fb \in I$. For a nonzero central λ the identity

$$gb\lambda = (f + \lambda g)b - fb$$

places $I + gbR$ inside $I + (f + \lambda g)R$, and a generic λ makes the dimension drop (Section 5). The infinite centre enters only through generic central scalars, in the choice of H (Lemma 4.3) and of λ, and the finiteness of $\operatorname{Kdim}(R/I)$ only through the termination of the induction.

We do not know whether Theorem A holds without (Z), nor whether Theorem A′ holds when $\operatorname{Kdim}(R/I)$ is infinite.

2 Preliminaries

Rings are associative with identity, and R^{op} denotes the opposite ring; a left R-module is a right R^{op}-module. A right ideal of R is *essential* if it meets every nonzero right ideal nontrivially.

Krull dimension

We use the Krull dimension of a module only through integer bounds and the bound ω. The zero module has Krull dimension -1; for an integer $n \geq 0$, $\operatorname{Kdim} M \leq n$ means that for every descending chain $M_0 \supseteq M_1 \supseteq \cdots$ of submodules all but finitely many factors M_i/M_{i+1} have Krull dimension at most $n-1$; and $\operatorname{Kdim} M \leq \omega$ means that for every such chain all but finitely many factors have finite Krull dimension. This is the deviation of the submodule lattice of M in the sense of [5, 6.1.2, 6.2.2]. Noetherian modules have a Krull dimension, and $\operatorname{Kdim} M = \max(\operatorname{Kdim} M', \operatorname{Kdim} M'')$ for every exact sequence $0 \to M' \to M \to M'' \to 0$ [5, 6.2.3, 6.2.4].

Fix an integer $e \geq 0$. A module M is *e-small* if $\operatorname{Kdim} M < e$, that is, $\operatorname{Kdim} M \leq e-1$; thus a module is 0-small exactly when it is zero. By the exact-sequence formula, e-small modules are closed under submodules, quotients, extensions and finite sums. A submodule $U \subseteq M$ is *e-dense* if M/U is e-small. A module C is *e-critical* if $\operatorname{Kdim} C = e$ and C/C' is e-small for every nonzero submodule C' [5, 6.2.9]. Every nonzero submodule of an e-critical module is e-critical [5, 6.2.11], and an e-critical module C has no nonzero e-small submodule C', since otherwise C would be an extension of the e-small modules C' and C/C'.

Proposition 2.1 ([5, 6.2.10]) *Every nonzero module M with $\operatorname{Kdim} M \leq n$ for an integer n has an e-critical submodule for some integer $e \leq n$. Consequently, a Noetherian module M with $\operatorname{Kdim} M \leq n$ has a chain $0 = M_0 \subset M_1 \subset \cdots \subset M_l = M$ of submodules whose factors M_j/M_{j-1} are e_j-critical for integers $e_j \leq n$.*

The consequence follows by applying the first statement to M/M_{j-1} in turn; the chain stops because M is Noetherian. We call such a chain a *critical series* of M, and, for $n = e$, its e-critical factors its *top factors*.

Goldie's theorem

A simple right Noetherian ring is prime and right Noetherian, so Goldie's theorem applies to it. We use it in the following form.

Theorem 2.2 (Goldie [23, Thms. 4.1, 4.4], [5, 2.3.5, 2.3.6]) *Let R be a simple right Noetherian ring.*

(i) *A right ideal of R is essential if and only if it contains a regular element.*

(ii) *The regular elements of R satisfy the right Ore condition: for every $a \in R$ and every regular $s \in R$ there are $a' \in R$ and a regular $s' \in R$ with $as' = sa'$.*

Two consequences are used repeatedly. If a right ideal I contains a regular element s, then for every $f \in R$ part (ii) gives a regular b with $fb \in sR \subseteq I$; in particular R/I is Goldie-torsion. And finitely many elements $y_1, \dots, y_n$ of a Goldie-torsion module have a common regular annihilator: each right ideal $\{r : y_i r = 0\}$ contains a regular element, hence is essential by (i); a finite intersection of essential right ideals is essential, and so contains a regular element by (i).

Lemma 2.3 ([5, 6.2.4, 6.3.9]) *Let R be a right Noetherian ring with $\operatorname{Kdim} R_R \leq \omega$. Then $\operatorname{Kdim}(R/I)$ is finite for every right ideal I that contains a regular element. In particular R satisfies* ($\mathrm{H}_{\mathrm{all}}$).

Indeed, for a regular $c \in I$ we have $\operatorname{Kdim}(R/cR) < \operatorname{Kdim} R_R \leq \omega$ by [5, 6.3.9], and R/I is a quotient of R/cR [5, 6.2.4].

Under ($\mathrm{H}_{\mathrm{all}}$) every finitely generated Goldie-torsion module has finite Krull dimension, being a quotient of $\bigoplus_i R/s_iR$ for regular s_i annihilating a set of generators.

3 Germs

Throughout this section R is right Noetherian and $e \geq 0$ is an integer. All modules in this section are finitely generated, hence Noetherian. Germs of level e are homomorphisms defined on e-dense submodules and taken modulo e-small modules. They play the role of the morphisms in the quotient of the module category by the modules of Krull dimension below e, in the sense of Gabriel [24, Ch. III], which Robson used to construct generators [14]; we work with them directly.

Definitions

For a module M let $\tau_e(M)$ be the sum of all e-small submodules of M. As M is Noetherian, it is a finite sum of them and so is e-small, and $\overline{M} = M/\tau_e(M)$ has no nonzero e-small submodule. A homomorphism $f\colon M \to N$ maps $\tau_e(M)$ into $\tau_e(N)$ and so induces $\overline{f}\colon \overline{M} \to \overline{N}$. An e-critical module C satisfies $\tau_e(C) = 0$.

Definition 3.1 Let M and N be modules. A *germ of level e* from M to N is an equivalence class $[U, \varphi]$ of homomorphisms $\varphi\colon U \to \overline{N}$ defined on e-dense submodules $U \subseteq \overline{M}$, where (U, φ) and (U', φ') are equivalent if φ and φ' agree on some e-dense submodule contained in $U \cap U'$. The germs form an abelian group $\operatorname{Hom}_e(M, N)$. The composite of $[V, \psi] \in \operatorname{Hom}_e(N, P)$ and $[U, \varphi] \in \operatorname{Hom}_e(M, N)$ is $[\varphi^{-1}(V), \psi\varphi]$. A germ $[U, \varphi]$ is *epic* if $\overline{N}/\varphi(U)$ is e-small.

These notions are well defined: $\varphi^{-1}(V)$ is e-dense because $U/\varphi^{-1}(V)$ embeds in $\overline{N}/V$, and if φ and φ' agree on an e-dense W, then $\varphi(U)/\varphi(W)$ is a quotient of the e-small module U/W, so $\overline{N}/\varphi(U)$ is e-small if and only if $\overline{N}/\varphi(W)$ is. Composition is associative and biadditive. A homomorphism $f\colon M \to N$ defines the germ $[\overline{M}, \overline{f}]$, again written f; it is zero if and only if $f(M)$ is e-small, and it is epic if and only if $N/f(M)$ is e-small, since $\overline{N}/\overline{f}(\overline{M}) \cong N/(f(M) + \tau_e(N))$ and $\tau_e(N)$ is e-small. Since $\overline{N}$ has no nonzero e-small submodule, a homomorphism $\varphi\colon U \to \overline{N}$ defined on an

e-dense submodule $U \subseteq M$ vanishes on $U \cap \tau_e(M)$ and so defines a germ from M to N, and every germ arises in this way; we use such representatives without further comment. A germ is *invertible* if it has a two-sided inverse under composition.

Sections. For a module N put $\Gamma_e(N) = \mathrm{Hom}_e(R, N)$. For $r \in R$ let $\ell_r \colon R \to R$, $y \mapsto ry$, which is a homomorphism of right modules, and define $w \cdot r = w \circ \ell_r$ for $w \in \Gamma_e(N)$. As $\ell_{rs} = \ell_r \ell_s$, this makes $\Gamma_e(N)$ a right R-module, and $h \circ (w \cdot r) = (h \circ w) \cdot r$ for every germ h from N to another module. Every $n \in N$ defines the section $\iota(n) \in \Gamma_e(N)$ given by $y \mapsto ny$, and $\iota(n) \cdot r = \iota(nr)$. For a right ideal J we write $q_J = \iota(1 + J) \in \Gamma_e(R/J)$; it is the germ of the projection $R \to R/J$.

Endomorphism germs. For an e-critical module C let $E_C = \mathrm{Hom}_e(C, C)$, a ring under composition. It acts from the left on $\mathrm{Hom}_e(M, C)$ and on $\Gamma_e(C)$ by composition, and on $\Gamma_e(C)$ this action commutes with the right action of R.

Lemma 3.2 (unit germs) *Let J be a right ideal of R and $r \in R$. The germ $q_J \cdot r$ of R/J is epic if and only if* $\mathrm{Kdim}\, R/(J + rR) < e$. *In particular q_J is epic.*

Proof The germ $q_J \cdot r$ is the germ of the homomorphism $R \to R/J$, $y \mapsto ry + J$, whose image is $(rR + J)/J$ and whose cokernel is $R/(J + rR)$. □

Lemma 3.3 (critical germs) (a) *Every nonzero germ of level e between e-critical modules is invertible. In particular E_C is a division ring for every e-critical module C.*

(b) *Let M be a module with* $\mathrm{Kdim}\, M \leq e$ *and a critical series, and let C be e-critical. Then* $\mathrm{Hom}_e(M, C)$ *is a left E_C-vector space whose dimension is at most the number of top factors of the series; and if* $\mathrm{Hom}_e(M, C) \neq 0$, *then C is isomorphic, through an invertible germ, to one of the top factors.*

Proof (a) Let C, D be e-critical and $[U, \varphi]$ a nonzero germ from C to D, with $U \subseteq C$ e-dense. Then $\varphi \neq 0$, so $U \neq 0$ is e-critical. If $\ker \varphi \neq 0$, then $\varphi(U) \cong U/\ker \varphi$ would be a nonzero e-small submodule of D; hence φ is injective. As $\varphi(U) \neq 0$, it is e-dense in D, and $[\varphi(U), \varphi^{-1}]$ is a two-sided inverse of $[U, \varphi]$. The identity germ of $C \neq 0$ is nonzero, so E_C is a division ring.

(b) We induct on the length l of the series; for $l = 0$ we have $M = 0$. Let $U = M_{l-1}$ and $F = M/U$. Restriction $\rho([V, \varphi]) = [V \cap U, \varphi]$ is an E_C-linear map $\mathrm{Hom}_e(M, C) \to \mathrm{Hom}_e(U, C)$; here $V \cap U$ is e-dense in U because $U/(V \cap U)$ embeds in M/V. If $\rho([V, \varphi]) = 0$, then φ vanishes on an e-dense submodule W of U, so $\varphi(V \cap U)$ is a quotient of the e-small module $(V \cap U)/W$ and hence zero, as C has no nonzero e-small submodule. Then φ induces $\varphi' \colon (V + U)/U \to C$ on the e-dense submodule $(V + U)/U$ of F, and $[V, \varphi] \mapsto [(V + U)/U, \varphi']$ is an injective E_C-linear map $\ker \rho \to \mathrm{Hom}_e(F, C)$ (it is well defined, since representatives that agree on an e-dense V_3 induce maps that agree on the e-dense $(V_3 + U)/U$): if φ' vanishes on an e-dense W'/U, then φ vanishes on $V \cap W'$, which is e-dense in M. Hence

$$\dim \mathrm{Hom}_e(M, C) \leq \dim \mathrm{Hom}_e(U, C) + \dim \mathrm{Hom}_e(F, C),$$

and $\mathrm{Hom}_e(M, C) \neq 0$ implies $\mathrm{Hom}_e(U, C) \neq 0$ or $\mathrm{Hom}_e(F, C) \neq 0$. If F is e-small, then $\overline{F} = 0$ and $\mathrm{Hom}_e(F, C) = 0$. If F is e-critical and $y \in \mathrm{Hom}_e(F, C)$ is nonzero, then y is invertible by (a), every $z \in \mathrm{Hom}_e(F, C)$ equals $(z \circ y^{-1}) \circ y$, and so $\dim \mathrm{Hom}_e(F, C) = 1$. The induction hypothesis for U completes the proof. □

Lemma 3.4 (epic criterion) *Let M be a module with $\operatorname{Kdim} M \le e$, with a critical series whose top factors are $C_1, \dots, C_m$, and let $x \in \operatorname{Hom}_e(N, M)$. Then x is epic if and only if for each i the map $\operatorname{Hom}_e(M, C_i) \to \operatorname{Hom}_e(N, C_i)$, $h \mapsto h \circ x$, is injective.*

Proof For the forward direction we show that a germ killed by x vanishes on an e-dense submodule; for the converse we build an e-critical quotient of $\overline{M}$ that x misses and map it to a top factor. Write $x = [U, \varphi]$. Suppose x is epic and $h \circ x = 0$ for $h = [V, \psi]$. Then $\psi\varphi$ vanishes on an e-dense $W \subseteq \varphi^{-1}(V)$, so ψ vanishes on $\varphi(W)$. This submodule is e-dense in $\overline{M}$, because $\overline{M}/\varphi(U)$ is e-small and $\varphi(U)/\varphi(W)$ is a quotient of U/W; hence $h = 0$.

Suppose x is not epic. Among the submodules $P \supseteq \varphi(U)$ of $\overline{M}$ with $\overline{M}/P$ not e-small choose a maximal one, which is possible as $\overline{M}$ is Noetherian. Then $Q' = \overline{M}/P$ is e-critical: $\operatorname{Kdim} Q' \le e$, Q' is not e-small, and Q'/Q'' is e-small for every nonzero Q'' by maximality. The projection $p\colon M \to Q'$ is a nonzero germ, as its image Q' is not e-small, and $p \circ x = 0$ since p vanishes on $\varphi(U)$. By Lemma 3.3(b) there is an invertible germ $a\colon Q' \to C_i$ for some i, and $h = a \circ p$ is a nonzero element of $\operatorname{Hom}_e(M, C_i)$ with $h \circ x = 0$. □

Lemma 3.5 (central scalars) *Let C be e-critical and λ in the centre $Z(R)$ of R. Right multiplication by λ is an endomorphism of C; let $\phi_\lambda \in E_C$ be its germ. Then ϕ_λ lies in the centre of E_C, the map $\lambda \mapsto \phi_\lambda$ is a ring homomorphism $Z(R) \to Z(E_C)$, and*

$$w \cdot \lambda = \phi_\lambda \circ w \qquad \text{for all } w \in \Gamma_e(C).$$

Proof If $\psi\colon V \to C$ represents a germ in E_C, then $V\lambda \subseteq V$ and $\psi(v\lambda) = \psi(v)\lambda$, so ψ commutes with ϕ_λ. The map $\lambda \mapsto \phi_\lambda$ is multiplicative and additive. If $\psi\colon U \to C$ represents w with $U \subseteq R$ a right ideal, then for $y \in U$ we have $\lambda y = y\lambda \in U$ and $\psi(\lambda y) = \psi(y\lambda) = \psi(y)\lambda$, so $w \cdot \lambda$ and $\phi_\lambda \circ w$ agree on U. □

If R is simple, $Z(R)$ is a field and $E_C \neq 0$, so $\lambda \mapsto \phi_\lambda$ is injective.

Lemma 3.6 (pencils) *Let E be a division ring, $\mathbb{F}$ a field and $\kappa\colon \mathbb{F} \to E$ a ring homomorphism with central image. Let D, W be left E-vector spaces with $\dim_E D < \infty$ and $P, P'\colon D \to W$ be E-linear, and call the family of E-linear maps $P + \kappa(\lambda)P'$, $\lambda \in \mathbb{F}$, a* pencil.

(i) *If P is injective, then $P + \kappa(\lambda)P'$ is injective for all but finitely many $\lambda \in \mathbb{F}$.*

(ii) *If P' is injective, then $P + \kappa(\lambda)P'$ is injective for all but finitely many $\lambda \in \mathbb{F}$.*

Proof (i) As E is a division ring and $\dim_E D < \infty$, P has an E-linear retraction $\pi\colon W \to D$. The value $\lambda = 0$ is good. If $\lambda \neq 0$ and $(P + \kappa(\lambda)P')(v) = 0$ with $v \neq 0$, then $\pi P'(v) = -\kappa(\lambda)^{-1}v$, so v is an eigenvector of $\pi P'$ for the central eigenvalue $-\kappa(\lambda)^{-1}$. Since κ is injective, distinct bad λ give distinct central eigenvalues, and eigenvectors for distinct central eigenvalues are linearly independent, by the usual argument with a shortest linear relation. Hence there are at most $\dim_E D$ bad values.

(ii) For $\lambda \neq 0$ we have $P + \kappa(\lambda)P' = \kappa(\lambda)\big(P' + \kappa(\lambda^{-1})P\big)$, and left multiplication by the central unit $\kappa(\lambda)$ is injective and E-linear; apply (i) to the pair (P', P). □

From now on in this section R is simple and right Noetherian, and an *E-bimodule* is a right R-module W with a left action of a division ring E that commutes with the action of R.

Lemma 3.7 (escape) *Let W be a Goldie-torsion E-bimodule, $0 \neq g \in W$, $0 \neq b \in R$ and $Y \subseteq W$ a finite-dimensional E-subspace. Then $grb \notin Y$ for some $r \in R$.*

Proof A basis of Y has a common regular annihilator c by the remark after Theorem 2.2, and $Yc = 0$ because the actions commute. If $gRb \subseteq Y$, then $gRbc = 0$. The set $T = \{y \in R : gRy = 0\}$ is a two-sided ideal, proper because $g \neq 0$, hence zero by simplicity. But $bc \in T$ and $bc \neq 0$ as c is regular. □

Lemma 3.8 (maximal rank) *Let $\mathbb{F} \subseteq Z(R)$ be an infinite subfield, W a Goldie-torsion E-bimodule and $\kappa\colon \mathbb{F} \to E$ a ring homomorphism with central image such that $w\lambda = \kappa(\lambda)w$ for $\lambda \in \mathbb{F}$ and $w \in W$. Let $P\colon D \to W$ be an injective E-linear map with $\dim_E D < \infty$ and let $0 \neq b \in R$. Then there is $r \in R$ such that $v \mapsto P(v)\,rb$ is injective.*

Proof We show that a map $P_r\colon v \mapsto P(v)rb$ of maximal rank is injective. Each P_r is E-linear of rank at most $\dim_E D$; choose r_0 with $\operatorname{rank} P_{r_0}$ maximal, put $Y_0 = P_{r_0}(D)$, and suppose $0 \neq v_0 \in \ker P_{r_0}$. By Lemma 3.7 with $g = P(v_0)$ there is r_1 with $w = P(v_0)r_1b \notin Y_0$. Let $j\colon Y_0 \to D$ be an E-linear section of P_{r_0}. For $\lambda \in \mathbb{F}^\times$ the commuting actions give

$$P_{r_0+\lambda r_1}\big(j(y) + \mu\,\kappa(\lambda)^{-1}v_0\big) = y + \mu w + \kappa(\lambda)\,P(j(y))\,r_1b \qquad (y \in Y_0,\ \mu \in E).$$

The map $(y, \mu) \mapsto y + \mu w$ on $Y_0 \oplus E$ is injective, so by Lemma 3.6(i) the right side is injective in (y, μ) for some $\lambda \neq 0$, as $\mathbb{F}$ is infinite. Then $\operatorname{rank} P_{r_0+\lambda r_1} \geq \dim Y_0 + 1$, a contradiction. □

Lemma 3.9 (sections of torsion modules) *If C is an e-critical Goldie-torsion module, then $\Gamma_e(C)$ is Goldie-torsion.*

Proof By Goldie's theorem it suffices to show that the right ideal $\mathfrak{a} = \{y \in R : w \cdot y = 0\}$ of a section w is essential, and for this that it meets every yR with $y \neq 0$, since every nonzero right ideal contains such a yR. Let $0 \neq y \in R$; we may assume $w \cdot y \neq 0$. Represent $w \cdot y$ by $\psi\colon U \to C$ with $U \subseteq R$ an e-dense right ideal; as $\psi \neq 0$, there is $i \in U$ with $c = \psi(i) \neq 0$, and $(w \cdot y) \cdot i = \iota(c)$ since both are represented by $z \mapsto \psi(iz) = cz$. The map ι is injective, because $cR \neq 0$ is not e-small; so $yi \neq 0$. Choose a regular s with $cs = 0$. Then $w \cdot (yis) = \iota(cs) = 0$, so $yis \in \mathfrak{a} \cap yR$, and $yis \neq 0$ as s is regular. □

4 Choosing the element $x + Ht$

The induction of Section 5 carries finitely many conditions of the following kind.

Definition 4.1 A *test* is a tuple (e, V, q, z, c) in which $e \geq 0$ is an integer, V a finitely generated Goldie-torsion module with $\operatorname{Kdim} V \leq e$, q and z sections in $\Gamma_e(V)$ with q epic, and $c \in R$ regular. An element $f \in R$ *satisfies* the test if the section $z + q \cdot fc$ is epic.

For a right ideal J with R/J Goldie-torsion and $\operatorname{Kdim} R/J \le e$, the *unit test* of R/J is $(e, R/J, q_J, 0, 1)$; by Lemma 3.2, f satisfies it if and only if $\operatorname{Kdim} R/(J+fR) < e$.

Lemma 4.2 (stability) *Let R be simple and right Noetherian with centre k.*

(i) *If M is finitely generated with $\operatorname{Kdim} M \le e$, $v \in \Gamma_e(M)$ is epic and $u \in \Gamma_e(M)$, then $v + u\lambda$ is epic for all but finitely many $\lambda \in k$.*

(ii) *If f satisfies a test and $g \in R$, then $f + \lambda g$ satisfies it for all but finitely many $\lambda \in k$.*

Proof (i) Choose a critical series of M with top factors C_i. For $h \in \operatorname{Hom}_e(M, C_i)$, Lemma 3.5 gives $h \circ (v + u\lambda) = h \circ v + \phi_\lambda \circ (h \circ u)$, so the map $h \mapsto h \circ (v + u\lambda)$ is the pencil $P + \phi_\lambda P'$ of the E_{C_i}-linear maps $P(h) = h \circ v$ and $P'(h) = h \circ u$ into $\Gamma_e(C_i)$. Here $\dim \operatorname{Hom}_e(M, C_i) < \infty$ and E_{C_i} is a division ring by Lemma 3.3, and P is injective by Lemma 3.4. By Lemma 3.6(i) each pencil is injective for all but finitely many λ, and for the remaining λ Lemma 3.4 shows that $v + u\lambda$ is epic.

(ii) For a test (e, V, q, z, c), as λ is central, $z + q \cdot (f + \lambda g)c = (z + q \cdot fc) + (q \cdot gc)\lambda$; apply (i). □

Lemma 4.3 (simultaneous tests) *Assume* (S), (N) *and* (Z). *For every $x \in R$, every nonzero $t \in R$ and every finite list of tests there is $H \in R$ such that $x + Ht$ satisfies all of them.*

For a single test the conclusion is close to the left-right mirror of [19, Thm. 5.1], which is stated there for simple left Noetherian rings and has no hypothesis on the centre; the central scalars serve to satisfy several tests at once.

Proof We reduce each test to finitely many injectivity conditions on finite-dimensional spaces and satisfy them all at once, first for the factor Ht alone by maximal rank and pencils, then for $x + Ht$ by one more pencil. Let $k = Z(R)$ and let the tests be $(e_i, V_i, q_i, z_i, c_i)$. Choose a critical series of each V_i with $\operatorname{Kdim} \le e_i$, and let $\alpha = (i, C)$ run over the pairs of a test and a top factor C of that series. Put $E_\alpha = E_C$, $D_\alpha = \operatorname{Hom}_{e_i}(V_i, C)$, $W_\alpha = \Gamma_{e_i}(C)$, $b_\alpha = tc_i \neq 0$, and let $P_\alpha, Z_\alpha \colon D_\alpha \to W_\alpha$ be $P_\alpha(h) = h \circ q_i$ and $Z_\alpha(h) = h \circ z_i$. Then D_α is finite-dimensional over the division ring E_α (Lemma 3.3), P_α is injective (Lemma 3.4), W_α is a Goldie-torsion E_α-bimodule (Lemma 3.9; C is a subquotient of V_i), and $w\lambda = \phi_\lambda w$ on W_α with $\lambda \mapsto \phi_\lambda$ injective with central image (Lemma 3.5). For $H \in R$,

$$h \circ \big(z_i + q_i \cdot (x + Ht)c_i\big) = X_\alpha(h) + P_\alpha(h)\, Hb_\alpha, \qquad X_\alpha(h) = Z_\alpha(h) + P_\alpha(h)\, xc_i,$$

so by Lemma 3.4 the element $x + Ht$ satisfies all tests if every map $h \mapsto X_\alpha(h) + P_\alpha(h)Hb_\alpha$ is injective. For $r, r' \in R$ and $\lambda \in k$ we have $P_\alpha(h)(r + \lambda r')b_\alpha = P_\alpha(h)rb_\alpha + \phi_\lambda P_\alpha(h)r'b_\alpha$, as the actions commute.

Step 1. There is H_0 such that every map $h \mapsto P_\alpha(h)H_0b_\alpha$ is injective. We induct on the number of pairs. If H' works for some of the pairs and, by Lemma 3.8, H'' works for one more pair β, then for $H' + \lambda H''$ the map for each earlier pair is a pencil $P + \phi_\lambda P'$ with P injective and the map for β is a pencil with P' injective. By Lemma 3.6 all but finitely many λ make all of them injective, and k is infinite.

Step 2. For $H = \lambda H_0$ the maps $h \mapsto X_\alpha(h) + \phi_\lambda P_\alpha(h)H_0b_\alpha$ are pencils whose second map is injective by Step 1, so by Lemma 3.6(ii) all of them are injective for all but finitely many $\lambda \in k$. Choose such a λ. □

5 Proof of the main theorems

Proof of Theorem A′ We prove by downward induction on e a statement that asks in addition for finitely many tests; each step uses the inductive hypothesis twice for the same ideal I and then combines the two answers with a generic central scalar. Let $k = Z(R)$, let $s \in I$ be regular and choose an integer $N \geq 0$ with $\operatorname{Kdim}(R/I) \leq N$. For $0 \leq e \leq N+1$ let $\mathrm{P}(e)$ be the assertion:

> for every $x \in R$, every nonzero $t \in R$ and every finite list L of tests there is $H \in R$ such that $f = x + Ht$ satisfies every test in L and $\operatorname{Kdim} R/(I + fR) < e$.

$\mathrm{P}(N+1)$ holds by Lemma 4.3, since $R/(I+fR)$ is a quotient of R/I. We show that $\mathrm{P}(e+1)$ implies $\mathrm{P}(e)$ for $0 \leq e \leq N$; then $\mathrm{P}(0)$ holds, and $\mathrm{P}(0)$ with the empty list is Theorem A′, because $\operatorname{Kdim} R/(I+fR) < 0$ means $I + fR = R$. Fix x, t and L.

Step 1 (a first element f). By $\mathrm{P}(e+1)$ there is H such that $f = x + Ht$ satisfies L and $J = I + fR$ has $\operatorname{Kdim} R/J \leq e$.

Step 2 (torsion equation). By Goldie's theorem there is a regular b with $fb \in I$ (remark after Theorem 2.2). Then $tb \neq 0$.

Step 3 (a correction g for the same ideal). The module R/J is Goldie-torsion, as $s \in J$, and $\operatorname{Kdim} R/J \leq e$, so it has a unit test. Apply $\mathrm{P}(e+1)$ with $x = 0$, the nonzero element tb and the list consisting of the unit test of R/J. This gives H_2 such that, with $g = H_2 t$, the element $H_2 tb = gb$ satisfies

$$\operatorname{Kdim} R/(J + gbR) < e \qquad \text{and} \qquad \operatorname{Kdim} R/Q \leq e, \quad \text{where } Q = I + gbR. \tag{1}$$

Step 4 (an epic section modulo Q). Since $I \subseteq J$, we have $J + gbR = I + fR + gbR = Q + fR$. By (1) and Lemma 3.2 applied to Q, the section $v = q_Q \cdot f$ of R/Q is epic. Put $u = q_Q \cdot g$.

Step 5 (a central scalar). By Lemma 4.2, applied to $M = R/Q$ and to the tests in L, all but finitely many $\lambda \in k$ make $v + u\lambda$ epic and $f + \lambda g$ satisfy L. As k is infinite, choose such a $\lambda \neq 0$.

Step 6 (inclusion). Put $f' = f + \lambda g$ and $J' = I + f'R$. Since λ is central,

$$gb\lambda = f'b - fb \in J', \tag{2}$$

because $f'b \in f'R$ and $fb \in I$. As λ is a unit, $gb \in J'$, and hence $Q + f'R = J'$.

Step 7 (the dimension drops). As λ is central, $q_Q \cdot f' = v + u\lambda$, which is epic. By Lemma 3.2, $\operatorname{Kdim} R/(Q + f'R) < e$, that is, $\operatorname{Kdim} R/J' < e$.

Step 8 (affine form). $f' = x + (H + \lambda H_2)t$, and f' satisfies every test in L. This proves $\mathrm{P}(e)$. □

Proof of Theorem A Put $I = dR$, which contains the regular element d, and $x = Bd$, $t = ad$; then $t \neq 0$ because $a \neq 0$ and d is regular. Theorem A′ gives H with

$$R = dR + (Bd + Had)R = dR + (B + Ha)dR,$$

that is, $1 = dU + (B + Ha)dV$ for some $U, V \in R$. For $a = 1$ and $B = 0$ this is $R = dR + HdR$. □

Proof of Corollary B By Lemma 2.3, $\operatorname{Kdim}(R/I)$ is finite for every right ideal I containing a regular element, in particular for $I = dR$ with d regular. Theorems A′ and A apply. □

6 Modules and right ideals

Proposition 6.1 *Assume* (S), (N) *and* (Z).

(a) *Let c, u be Goldie-torsion elements of a module with* $\mathrm{Kdim}(uR)$ *finite. Then for every nonzero $t \in R$ there is $H \in R$ with $cR + uR = (c + uHt)R$.*

(b) *Let M be a finitely generated Goldie-torsion module with* $\mathrm{Kdim}\, M$ *finite. Then for every $x \in M$ and every nonzero $t \in R$ there is $h \in M$ with $M = (x + ht)R$.*

Proof (a) We apply Theorem A′ to the annihilator of u. Choose a regular b with $cb = 0$. The right ideal $I_1 = \{r \in R : ur = 0\}$ contains a regular element, and $R/I_1 \cong uR$ has finite Krull dimension. Theorem A′ with $x = 0$ and the nonzero element tb gives H and $r \in R$ with $1 - Htbr \in I_1$, so that $u = uHtbr$. For $z = c + uHt$ we get $zbr = cbr + uHtbr = u$, so $u \in zR$ and $c = z - uHt \in zR$. Hence $cR + uR = zR$.

(b) We induct on the number m of generators $u_1, \ldots, u_m$ of M; for $m = 0$ take $h = 0$. The module M/u_1R is generated by $m - 1$ elements and is Goldie-torsion with $\mathrm{Kdim}(M/u_1R) \leq \mathrm{Kdim}\, M$, so by induction there is $y \in M$ with $M = cR + u_1R$ for $c = x + yt$. By (a), as $\mathrm{Kdim}(u_1R) \leq \mathrm{Kdim}\, M$, there is H with $cR + u_1R = (c + u_1Ht)R$. Take $h = y + u_1H$. □

Proposition 6.2 *Let R be simple and right Noetherian, and I a right ideal of R.*

(a) *There is $a \in I$ such that every $y \in I$ has a regular s with $ys \in aR$.*

(b) *Assume moreover* (Z) *and* ($\mathrm{H}_{\mathrm{all}}$). *For every a as in (a), every $y \in I$ and every nonzero $t \in R$ there is $h \in I$ with $I = aR + (y + ht)R$. In particular I is generated by two elements.*

Proof (a) We take for a the I-component of a regular element of the essential right ideal $I \oplus C$. Choose a right ideal C maximal with $I \cap C = 0$, which exists as R is right Noetherian. The right ideal $I \oplus C$ is essential, since $K \cap (I \oplus C) = 0$ for a right ideal K would give $I \cap (C + K) = 0$ and hence $K = 0$ by the maximality of C; so by Goldie's theorem it contains a regular element $s = a + c$ with $a \in I$, $c \in C$, and sR is essential. Let $y \in I$. The right ideal $\{r \in R : yr \in sR\}$ is essential: if K is a nonzero right ideal, either $yK = 0 \subseteq sR$, or yK is a nonzero right ideal and meets sR in some $yr \neq 0$ with $r \in K$. So it contains a regular element s', and $ys' = sr' = ar' + cr'$ for some r'. Then $cr' = ys' - ar' \in I \cap C = 0$, so $ys' = ar' \in aR$.

(b) The module I/aR is finitely generated by (N) and Goldie-torsion by the choice of a. By ($\mathrm{H}_{\mathrm{all}}$) its Krull dimension is finite (Section 2). Proposition 6.1(b) applied to $y + aR$ gives $h \in I$ with $I/aR = (y + ht + aR)R$, that is, $I = aR + (y + ht)R$. For two generators take $y = 0$ and $t = 1$. □

Proof of Corollary C By Lemma 2.3, (H_ω) implies ($\mathrm{H}_{\mathrm{all}}$). Hence (C.1) is Proposition 6.1(b) by the remark after Lemma 2.3, and (C.2) is Proposition 6.2. □

7 Applications

The applications rest on three classes of rings: filtered algebras with commutative Noetherian associated graded ring, the Weyl algebras, and the rings of differential operators on smooth affine varieties. For a field k, an *exhaustive filtration* of a k-algebra

R is a chain $F_0R \subseteq F_1R \subseteq \cdots$ of k-subspaces with $1 \in F_0R$, $F_iR\,F_jR \subseteq F_{i+j}R$ and $\bigcup_i F_iR = R$; its associated graded ring is $\operatorname{gr} R = \bigoplus_i F_iR/F_{i-1}R$ with $F_{-1}R = 0$.

Theorem 7.1 ([5, 1.6.9, 6.5.6, 6.4.8]) *Let R be a k-algebra with an exhaustive filtration such that $\operatorname{gr} R$ is a commutative Noetherian ring of classical Krull dimension at most m. Then R is right Noetherian [5, 1.6.9] and $\operatorname{Kdim} R_R \leq m$ [5, 6.5.6, 6.4.8].*

The opposite filtration of R^{op} has $\operatorname{gr}(R^{\mathrm{op}}) \cong (\operatorname{gr} R)^{\mathrm{op}} = \operatorname{gr} R$, so the theorem applies to R^{op} as well.

For a field k of characteristic zero and $n \geq 0$, the Weyl algebra $A_n(k)$ is the k-algebra generated by $x_1, \ldots, x_n, \partial_1, \ldots, \partial_n$ with the relations $\partial_i x_j - x_j \partial_i = \delta_{ij}$ and all other pairs of generators commuting. The assignment $x_i \mapsto x_i$, $\partial_i \mapsto -\partial_i$ preserves these relations when products are reversed, and so defines an isomorphism $\theta\colon A_n(k) \to A_n(k)^{\mathrm{op}}$.

Theorem 7.2 ([5, 1.3.5, 6.6.15]) *Let k be a field of characteristic zero. Then $A_n(k)$ is a simple right Noetherian domain whose centre contains k and is therefore infinite, and $\operatorname{Kdim} A_n(k) \leq \omega$.*

Let k be a field of characteristic zero and A a smooth k-algebra of finite type that is a domain, the coordinate ring of a smooth irreducible affine variety X. The ring $D(X) = D_k(A)$ of differential operators is the subalgebra of $\operatorname{End}_k(A)$ of the operators of finite order, where an operator P has order at most 0 if it commutes with every multiplication by an element of A, and order at most $j+1$ if every commutator $[P, a]$ with $a \in A$ has order at most j; products are compositions.

Theorem 7.3 ([5, 15.1.20, 15.3.7, 15.4.5, 15.4.7, 15.5.6], [25, Thm. 3.7]) *The ring $D(X)$ is a simple right and left Noetherian domain whose centre contains k and is therefore infinite, and $\operatorname{Kdim} D(X)_{D(X)}$ and $\operatorname{Kdim}{}_{D(X)} D(X)$ are finite.*

Simplicity is [5, 15.3.7] and [25, Thm. 3.7]. For the Krull dimension, let A be generated by r elements $x_1, \ldots, x_r$. As A is smooth, there are derivations $E_1, \ldots, E_r$ of A with $\mathrm{d}b = \sum_i E_i(b)\,\mathrm{d}x_i$ in the module of Kähler differentials for every $b \in A$. In characteristic zero $D(X)$ is generated by A and $\operatorname{Der} A$ [5, 15.5.6], so the order filtration of $D(X)$ has an associated graded ring that is a quotient of $A[\xi_1, \ldots, \xi_r]$, with ξ_i the symbol of E_i. This is a commutative Noetherian ring of Krull dimension at most $2r$, so $\operatorname{Kdim} D(X) \leq 2r$ on both sides by Theorem 7.1.

In the following theorem a *torsion* module over a domain is one in which every element is annihilated by a nonzero element. The left forms of the identities are those of R^{op} written in R.

Theorem 7.4 (a) Filtered rings. *Let k be an infinite field and R a simple k-algebra with an exhaustive filtration such that* $\operatorname{gr} R$ *is commutative Noetherian of classical Krull dimension at most m. Then R and R^{op} satisfy* (N), (Z), $\operatorname{Kdim} R_R \le m$ *and* $\operatorname{Kdim} R^{\mathrm{op}}_{R^{\mathrm{op}}} \le m$. *Hence Theorems A' and A and Corollary C hold for R and for R^{op}, for every regular d and every right, respectively left, ideal containing a regular element; in particular $1 = Ud + Vd(B + aH)$ for some H, U, V, for every regular d, every nonzero a and every B. Moreover, for Goldie-torsion elements c, u of a right R-module and every nonzero t there is H with $cR + uR = (c + uHt)R$, and likewise for left modules.*

(b) Weyl algebras. *Let k be a field of characteristic zero and $A = A_n(k)$. For all nonzero $d, a \in A$ and every $B \in A$ there are H, U, V with $1 = dU + (B + Ha)dV$; in particular, for every nonzero d there are F, U, V with $1 = dU + FdV$, which is Stafford's Conjecture 3.8. For every nonzero right ideal I, every $x \in A$ and every nonzero t there is H with $I + (x + Ht)A = A$. Every finitely generated torsion right module M satisfies $M = (x + ht)A$, and every finitely generated torsion left module M satisfies $M = A(x + th)$, for given $x \in M$, $t \neq 0$ and some $h \in M$; in particular these modules are cyclic. If a is a nonzero element of a right ideal I, then for every $y \in I$ and nonzero t there is $h \in I$ with $I = aA + (y + ht)A$; similarly $I = Aa + A(y + th)$ for left ideals; in particular one-sided ideals of A are generated by two elements.*

(c) Differential operators. *Let $D = D(X)$ be as above. Every right and every left ideal of D is generated by two elements. For all nonzero $d, a \in D$ and every $B \in D$ there are $H, U, V \in D$ with $1 = dU + (B + Ha)dV$, and $H', U', V' \in D$ with $1 = U'd + V'd(B + aH')$; in particular, for every nonzero d there are $F, U, V \in D$ with $1 = dU + FdV$. Every finitely generated torsion right module M satisfies $M = (x + ht)D$, and every finitely generated torsion left module M satisfies $M = D(x + th)$, for given $x \in M$, $t \neq 0$ and some $h \in M$.*

(d) Enveloping algebras. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over an infinite field k and R a simple quotient of $U(\mathfrak{g})$. Then R and R^{op} satisfy* (N), (Z), $\operatorname{Kdim} R_R \le \dim \mathfrak{g}$ *and* $\operatorname{Kdim} R^{\mathrm{op}}_{R^{\mathrm{op}}} \le \dim \mathfrak{g}$, *and the conclusions of (a) hold for R and R^{op}.*

(e) Simple quantizations. *If in (a) the ring* $\operatorname{gr} R$ *is moreover a domain, then for all nonzero $d, a \in R$ and every $B \in R$ there are H, U, V with $1 = dU + (B + Ha)dV$, and H', U', V' with $1 = U'd + V'd(B + aH')$.*

Proof (a) By Theorem 7.1, applied to R and to R^{op} with the opposite filtration, both are right Noetherian with $\operatorname{Kdim} \le m$. The centre of R contains the image of k, which is infinite since $R \neq 0$, and R^{op} is simple with the same centre. Corollaries B and C and Proposition 6.1(a) apply to R and to R^{op}; an identity $1 = dU + (B + Ha)dV$ in R^{op} reads $1 = Ud + Vd(B + aH)$ in R.

(b) By Theorem 7.2, A satisfies (S), (N), (Z) and (H_ω). As A is a domain, nonzero elements are regular, torsion modules are Goldie-torsion, and a nonzero right ideal contains a regular element; Corollaries B and C apply. If $0 \neq a \in I$ and $y \in I$, Goldie's theorem (ii) gives a regular s with $ys \in aA$, so a satisfies the condition of (C.2). The isomorphism $\theta\colon A \to A^{\mathrm{op}}$ carries left ideals and left modules to right ideals and right modules, which gives the left forms.

(c) By Theorem 7.3, D and D^{op} satisfy (S), (N), (Z) and (H_ω), and D is a domain; argue as in (b), with D^{op} in place of θ.

(d) The standard filtration of $U(\mathfrak{g})$ induces an exhaustive filtration of R whose associated graded ring is a quotient of $\operatorname{gr} U(\mathfrak{g})$, which is commutative and generated by the images of

a basis of $\mathfrak{g}$ [5, 1.7.4]. So $\operatorname{gr} R$ is a quotient of a polynomial ring in $\dim \mathfrak{g}$ variables, hence commutative Noetherian of Krull dimension at most $\dim \mathfrak{g}$, and (a) applies.

(e) As $\operatorname{gr} R$ is a domain, so is R [5, 1.6.6], so nonzero elements are regular, and (a) applies. □

In the sense of Bellamy [3, §7, p. 12], a quantization of a conic symplectic singularity X is a filtered $\mathbb{C}$-algebra $U = \bigcup_{n\geq 0} F_n U$ with $\operatorname{gr} U \cong \mathbb{C}[X]$ as graded Poisson algebras, where X is a normal affine variety and $\mathbb{C}[X]$ is connected graded, so that X is irreducible. Hence $\operatorname{gr} U$ is a commutative Noetherian domain of finite Krull dimension, and a simple such U satisfies the hypotheses of parts (a) and (e). This answers, for the two-generation of one-sided ideals and the cyclicity of finitely generated Goldie-torsion modules, the question of [3, §7.1] whether the results of [1] hold for simple quantizations.

8 Formalization

Every numbered statement of this paper, including the results quoted from the literature, is formalized in Lean 4 [26] with Mathlib [27] and the library AlgebraicAnalysis [28], in the repository generalized-stafford-formal [29] at commit `dd9447b`, with AlgebraicAnalysis at commit `22b3946` as recorded in its lock file. The main statements, Theorems A′ and A, Corollaries B and C and Theorem 7.4, are stated in a separate file that imports only Mathlib and defines Krull dimension bounds and Goldie torsion explicitly; they are proved without unproved assumptions and use only the standard axioms. The tool Comparator [30] checks that the proved theorems have exactly the stated types, and the Lean kernel and the independent checker NanoDa [31] accept the proofs. To check, build the repository at that commit with `lake build` and run Comparator on the configuration file of the repository.

The Palomar registry records formal statements together with machine-checked proofs of them. Its entry [32] is a Lean proof of Stafford's Conjecture 3.8 for nonzero d; the identity $1 = dU + FdV$ of Theorem 7.4(b) is stated in Lean as that registered statement, with the definitions of $A_n(k)$ and of the identity copied verbatim. Its entry [33] is a Lean proof of the identity $1 = dU + FdV$ for every nonzero d in $D(X)$; the corresponding identity of Theorem 7.4(c) is stated in Lean with the same definitions of differential operators and the same proposition. The proofs given here are different.

AI-assisted research and formalization. AI agents, mainly OpenAI GPT Sol 6.1 and Anthropic Claude Opus 5.5, assisted with proof discovery, counterexample searches, review, formalization and drafting of the manuscript. Each proof was reviewed independently before it was formalized, and the Lean statements were checked against the text. The authors are responsible for the final text.

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